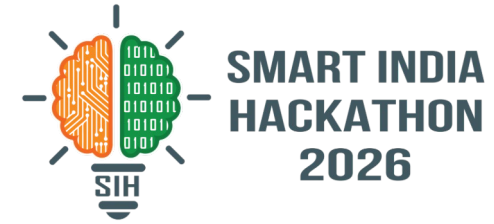
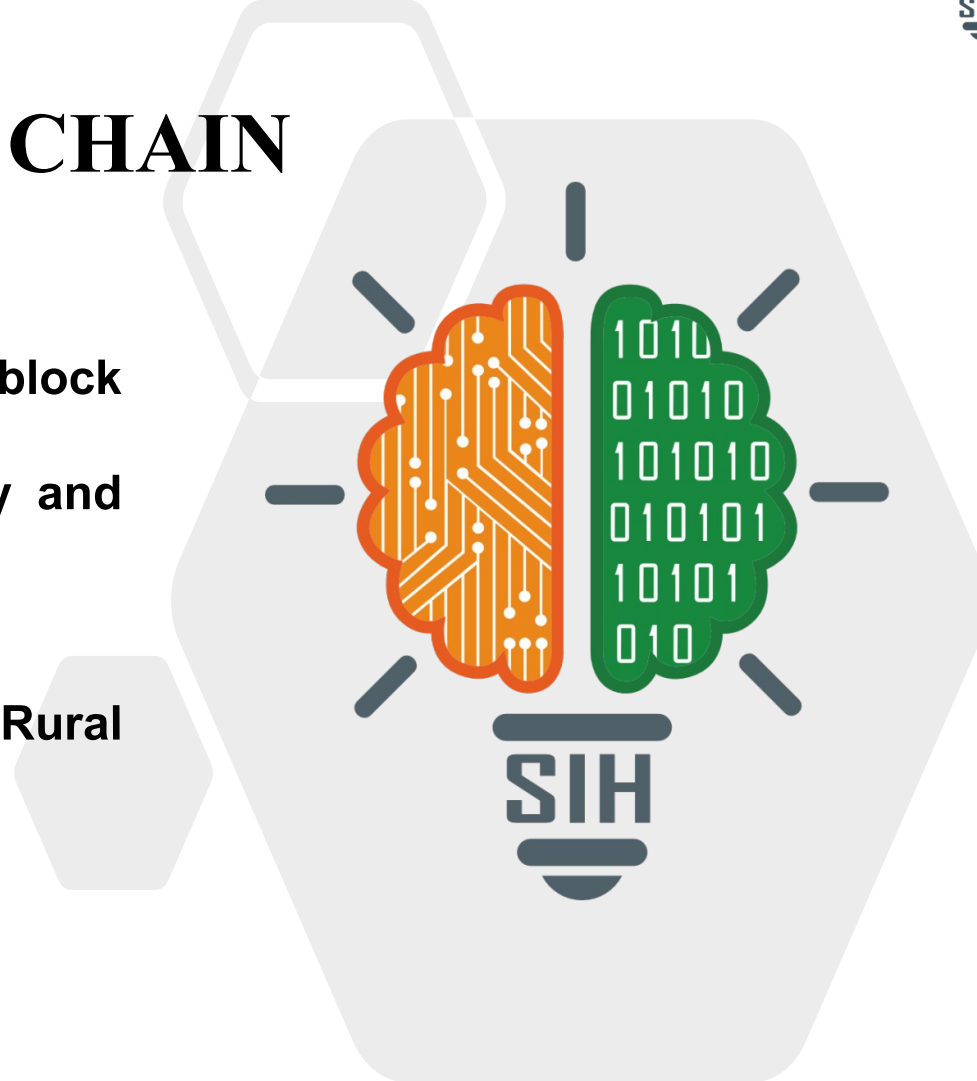


SMART INDIA HACKATHON 2026



HONEY CHAIN

- Problem Statement ID – SIH26021
- Problem Statement Title- Honey Chain:A block chain based system for honey traceability and smart beekeeping management
- Theme- Agriculture,Foodtech & Rural Development
- PS Category- Software
- Team ID-135807
- Team Name - Code_Catalyst01



Proposed Solution: from beehive to consumer, every honey batch is traceable and verified



1 Detailed explanation

- Unique QR code per honey batch shows its verified hive-to-shelf journey
- Blockchain keeps tamper-resistant batch records
- IoT smart hive and AI track hive health and predict honey yield
- Beekeeper dashboard, smart alerts and a direct marketplace

2 How it addresses the problem

- Counterfeit honey: tamper-resistant records stop false relabelling
- Consumer trust: one scan shows source, quality tests and packaging
- Weak market linkage: verified beekeepers sell directly to buyers
- Hive losses: early alerts for disease, heat and weight loss

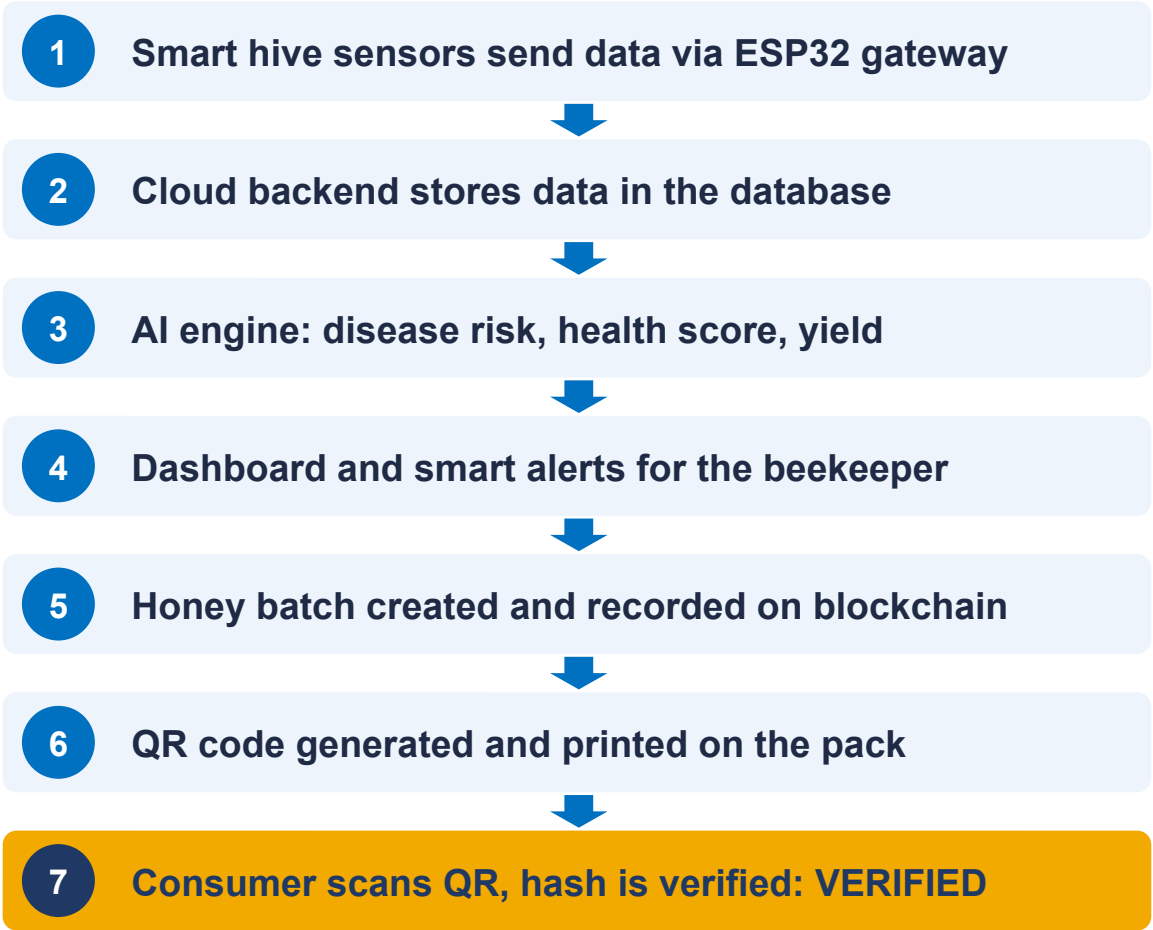
3 Innovation and uniqueness

- Four pillars in one platform: Trust, Transparency, Intelligence, Monitoring
- Starts at the hive, not at the packaging line
- QR is not the proof: backend checks the blockchain hash and flags suspicious scans
- Key events on-chain, bulky sensor data off-chain

Technologies to be used

Frontend	React.js, responsive web design (Flutter for mobile)
Backend	Python with FastAPI / Flask
Database	PostgreSQL / MySQL for app and IoT data
Blockchain	Ethereum-compatible chain, Solidity, Hardhat, MetaMask (or Hyperledger Fabric)
Smart contract	registerHive, createHoneyBatch, addQualityReport, transferBatch, verifyBatch
IoT	ESP32, DHT22, load cell + HX711, GPS, microphone
AI	Python, scikit-learn (TensorFlow / PyTorch for images)
QR	QR code generator and scanner

Methodology: implementation flow



Feasibility

- Low-cost, off-the-shelf sensors (ESP32, DHT22, HX711)
- Proven open-source stack: React, FastAPI, PostgreSQL, Solidity
- Only key events go on-chain; IoT data stays off-chain
- End-to-end demo possible with simulated hive data
- Marketplace can follow the core MVP



Challenges and risks

- Genuine QR codes copied onto fake bottles
- Storing every sensor reading on-chain is slow and costly
- Disease prediction needs enough quality hive data
- Trying to build every feature at once
- Ease of use for rural beekeepers



Strategies to overcome

- Verify via backend and blockchain hash; flag odd scan patterns
- Hybrid storage: blockchain plus PostgreSQL
- Start with sensor-based prediction; add image models later
- Focused 10-step MVP demo flow
- Simple mobile / web dashboard

Potential impact on the target audience

1 Beekeepers

Early hive alerts, yield forecasts and direct access to buyers

2 Consumers

Scan to verify origin, floral source and quality before buying

3 Buyers

Retailers, processors and restaurants source from verified batches

4 Honey industry

Tamper-resistant records discourage false labelling

Benefits of the solution

Social

- Empowers rural beekeepers with digital tools
- Restores consumer trust in honey
- Transparent, verifiable supply chain

Economic

- Fairer prices for verified honey
- Direct beekeeper-to-buyer sales
- Fewer colony and yield losses

Environmental

- Healthier colonies support pollination
- Inspect only when data says so: less disturbance
- Data-driven, sustainable beekeeping

Research basis

Hive monitoring

Temperature, humidity, weight and acoustic patterns indicate hive health

Blockchain traceability

Tamper-resistant batch records give supply-chain provenance

AI models

Disease-risk scoring and 14-day yield prediction with scikit-learn

Problem context

Counterfeit honey, low consumer trust and weak market linkage

Technical references

Ethereum developer docs

ethereum.org/developers

Solidity language

docs.soliditylang.org

Hardhat

hardhat.org

Hyperledger Fabric

hyperledger-fabric.readthedocs.io

FastAPI

fastapi.tiangolo.com

scikit-learn

scikit-learn.org

ESP32 (Espressif)

docs.espressif.com